Policy JSON.

 IAM policy: A document that defines which AWS actions are allowed or denied.

 JSON: A structured text format used to write IAM policies.

 Version: Identifies the IAM policy-language version. The current standard value is 2012-10-17.

 Statement: Contains one or more permission rules.

 Sid: An optional name used to identify a statement.

 Effect: Specifies whether access is Allow or Deny.

 Action: Specifies the AWS API operations, such as s3:GetObject.

 Resource: Specifies the AWS resource on which the action applies.

 ARN: Amazon Resource Name—a unique identifier for an AWS resource.

 Wildcard (*): Represents all applicable actions or resources.

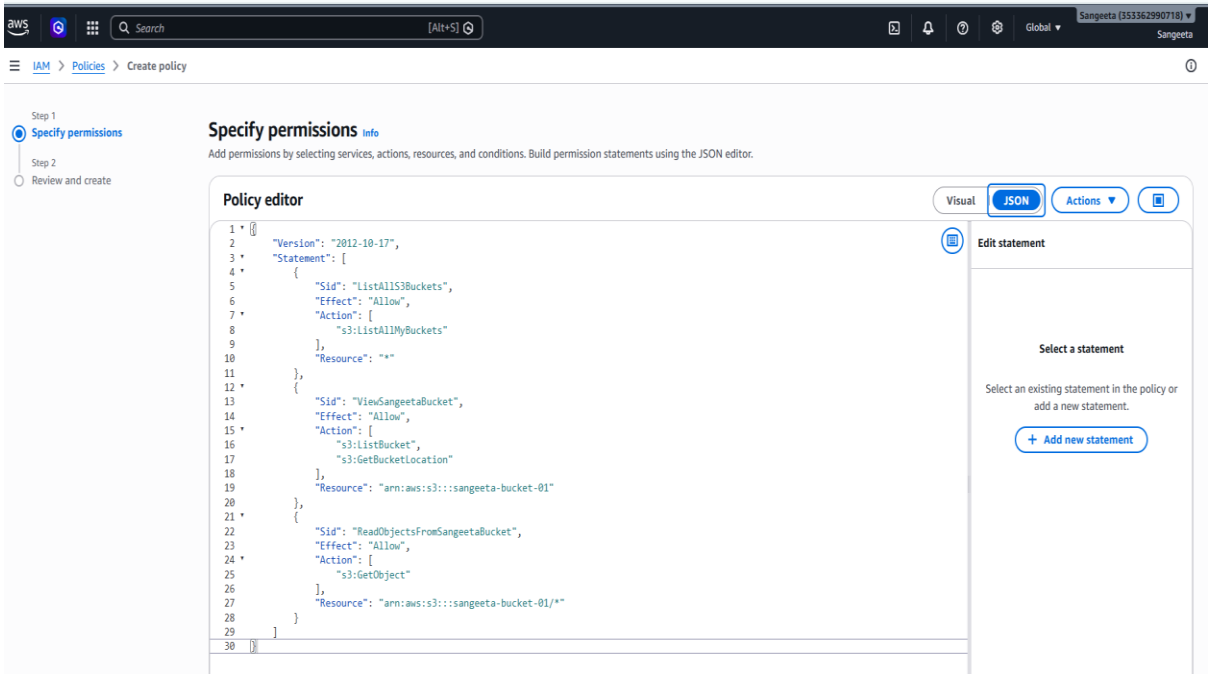
 Least privilege: Giving only the minimum permissions required to complete a task.

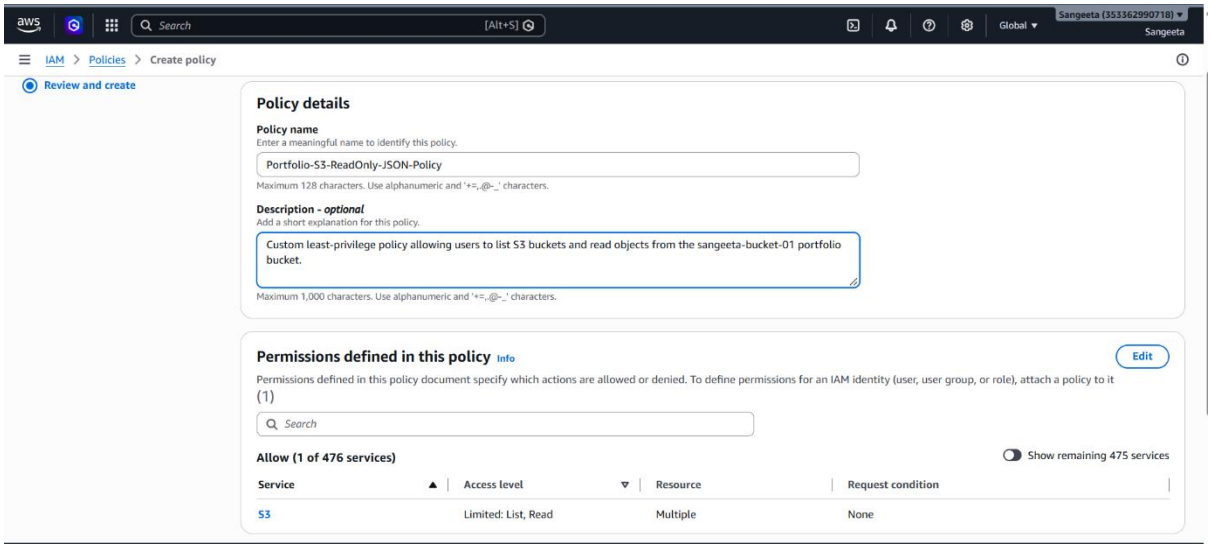
 Implicit deny: Access is denied automatically when no policy allows it.

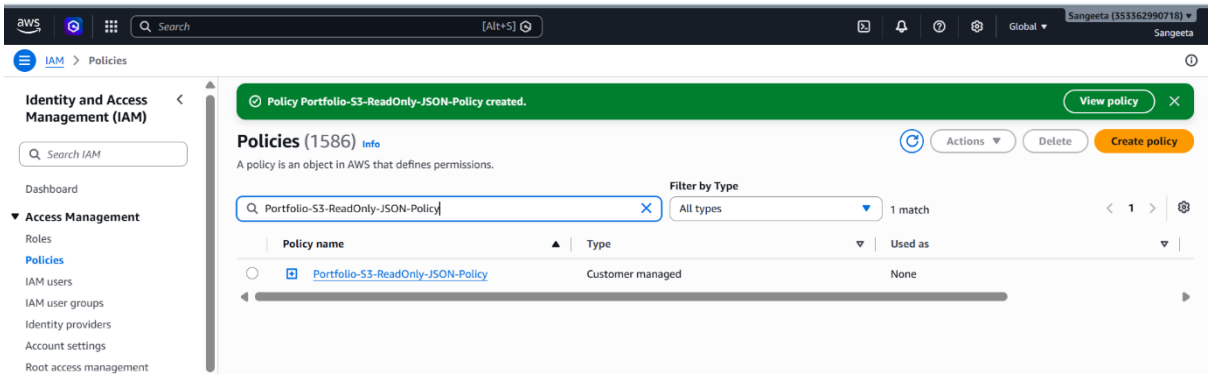
 Explicit deny: A policy specifically contains "Effect": "Deny"; it overrides an allow.

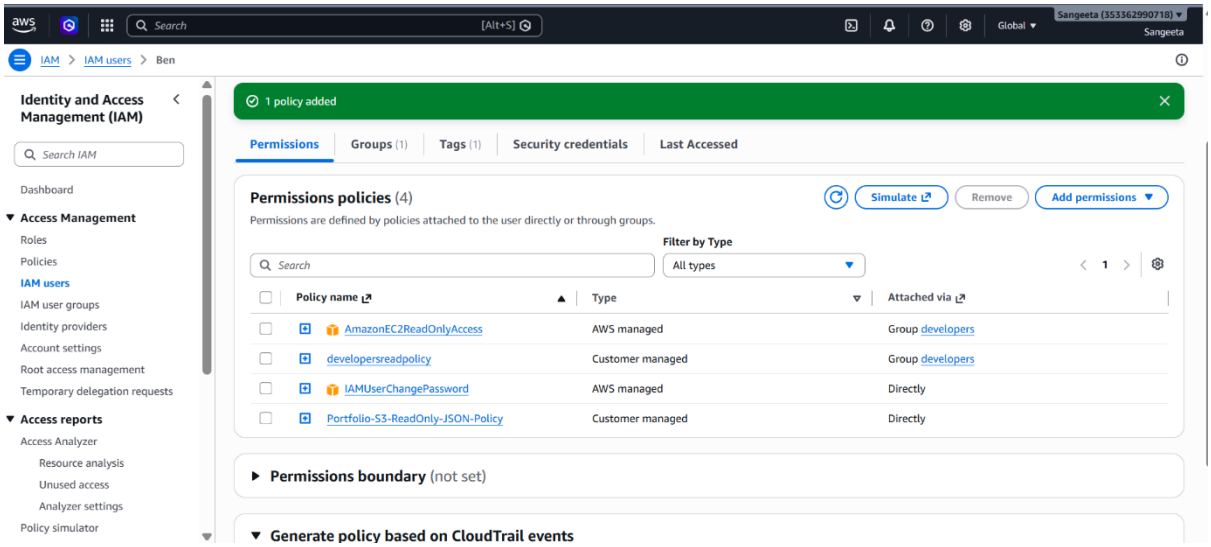
 Customer-managed policy: A reusable policy created and controlled within your AWS account.

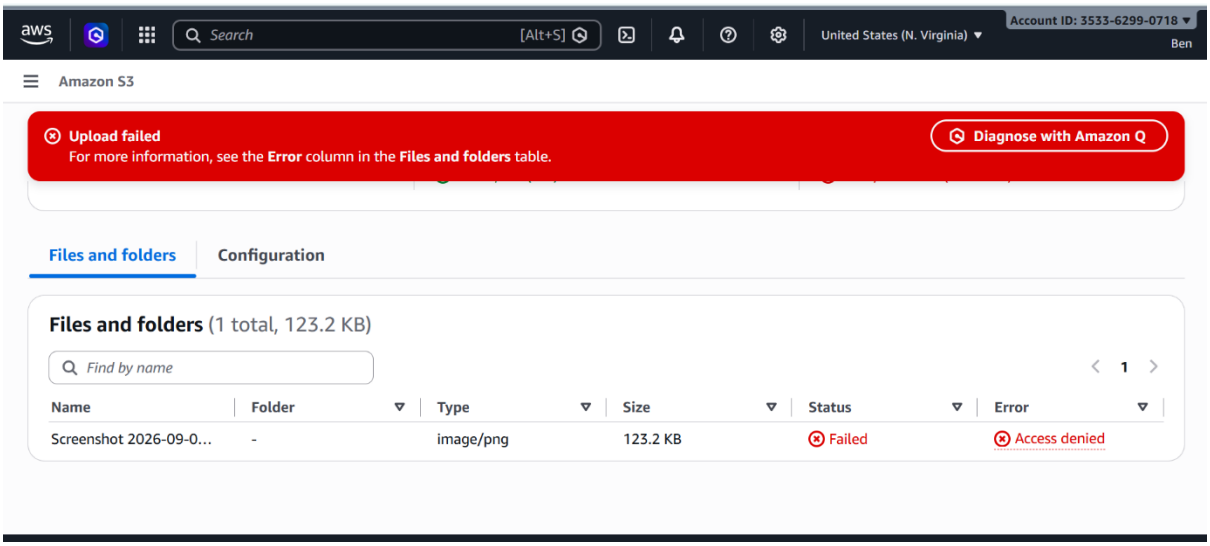
 AccessDenied: An error produced when the requested action is not permitted.






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Global
Sangeeta (353362990718)
Sangeeta
IAM > Policies
Identity and Access Management (IAM)
Search IAM
Dashboard
Access Management
Roles
Policies
IAM users
IAM user groups
Identity providers
Account settings
Root access management
Policy Portfolio-S3-ReadOnly-JSON-Policy created.
View policy
Policies (1586) Info
A policy is an object in AWS that defines permissions.
Actions
Delete
Create policy
Filter by Type
Portfolio-S3-ReadOnly-JSON-Policy
All types
1 match
Policy name
Type
Used as
Portfolio-S3-ReadOnly-JSON-Policy
Customer managed
None


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IAM > IAM users > Ben
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Account settings
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Temporary delegation requests
Access reports
Access Analyzer
Resource analysis
Unused access
Analyzer settings
Policy simulator
1 policy added
Permissions
Groups (1)
Tags (1)
Security credentials
Last Accessed
Permissions policies (4)
Permissions are defined by policies attached to the user directly or through groups.
Simulate
Remove
Add permissions
Filter by Type
Search
All types
Policy name
Type
Attached via
AmazonEC2ReadOnlyAccess
AWS managed
Group developers
developersreadpolicy
Customer managed
Group developers
IAMUserChangePassword
AWS managed
Directly
Portfolio-S3-ReadOnly-JSON-Policy
Customer managed
Directly
Permissions boundary (not set)
Generate policy based on CloudTrail events


Search [Alt+S]
United States (N. Virginia)
Account ID: 3533-6299-0718
Ben
Amazon S3
Upload failed
For more information, see the Error column in the Files and folders table.
Diagnose with Amazon Q
Files and folders
Configuration
Files and folders (1 total, 123.2 KB)
Find by name
Name
Folder
Type
Size
Status
Error
Screenshot 2026-09-0...
-
image/png
123.2 KB
Failed
Access denied

1. Project objective

To create a custom least-privilege IAM policy that allows an IAM user to:

- View the list of S3 buckets
- Open one specific bucket
- View and download its files
- Prevent uploading and deleting files

2. Resources used

- AWS IAM
- Amazon S3
- IAM user: Ben
- S3 bucket: sangeeta-bucket-01
- Custom policy: Portfolio-S3-ReadOnly-JSON-Policy

4. Implementation steps

Step 1: Open IAM Policies

1. Signed in to the AWS Management Console using the administrator account.
2. Opened **IAM — Identity and Access Management**.
3. Selected **Policies** from the left navigation menu.
4. Clicked **Create policy**.

Step 2: Open the JSON editor

1. Opened the **JSON** policy editor.
2. Removed the default JSON policy.
3. Entered a custom least-privilege policy.

Step 3: Add the JSON policy

4. The following policy was entered:

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "ListAllS3Buckets",
      "Effect": "Allow",
      "Action": [
        "s3:ListAllMyBuckets"
      ],
      "Resource": "*"
    },
    {
      "Sid": "ViewSangeetaBucket",
      "Effect": "Allow",
      "Action": [
```

```
      "s3:ListBucket",
      "s3:GetBucketLocation"
    ],
    "Resource": "arn:aws:s3:::sangeeta-bucket-01"
  },
  {
    "Sid": "ReadObjectsFromSangeetaBucket",
    "Effect": "Allow",
    "Action": [
      "s3:GetObject"
    ],
    "Resource": "arn:aws:s3:::sangeeta-bucket-01/*"
  }
 ]
}
```

Step 5: Create the policy

Portfolio-S3-ReadOnly-JSON-Policy

Step 6: Attach the policy to Ben

1. Opened **IAM → Users**.

2. Selected the user **Ben**.
3. Opened the **Permissions** tab.
4. Selected **Add permissions**.
5. Chose **Attach policies directly**.
6. Searched for Portfolio-S3-ReadOnly-JSON-Policy.
7. Selected the policy.
8. Clicked **Next** and **Add permissions**.

Step 7: Test allowed access

1. Opened Ben's IAM sign-in link in an Incognito/InPrivate browser.
2. Signed in using Ben's credentials.
3. Opened Amazon S3.
4. Opened sangeeta-bucket-01.
5. Confirmed that Ben could view the files.
6. Tested viewing or downloading an existing object.

Step 8: Test denied access

1. While signed in as Ben, opened sangeeta-bucket-01.
2. Selected **Upload**.
3. Attempted to upload a test file.

4. AWS denied the operation because the policy did not include s3:PutObject.

Final result

A customer-managed IAM policy was successfully created and attached to Ben. Ben received read-only access to sangeeta-bucket-01. Ben could view and download files but could not upload or delete objects.

Skills demonstrated

- AWS IAM user and policy management
- IAM JSON policy creation
- Amazon S3 permissions
- Resource-level ARN configuration
- Least-privilege access control
- Allowed and denied permission testing
- AccessDenied troubleshooting
- AWS security best practices

Policy Evaluation :

Part 1: Prepare two test objects

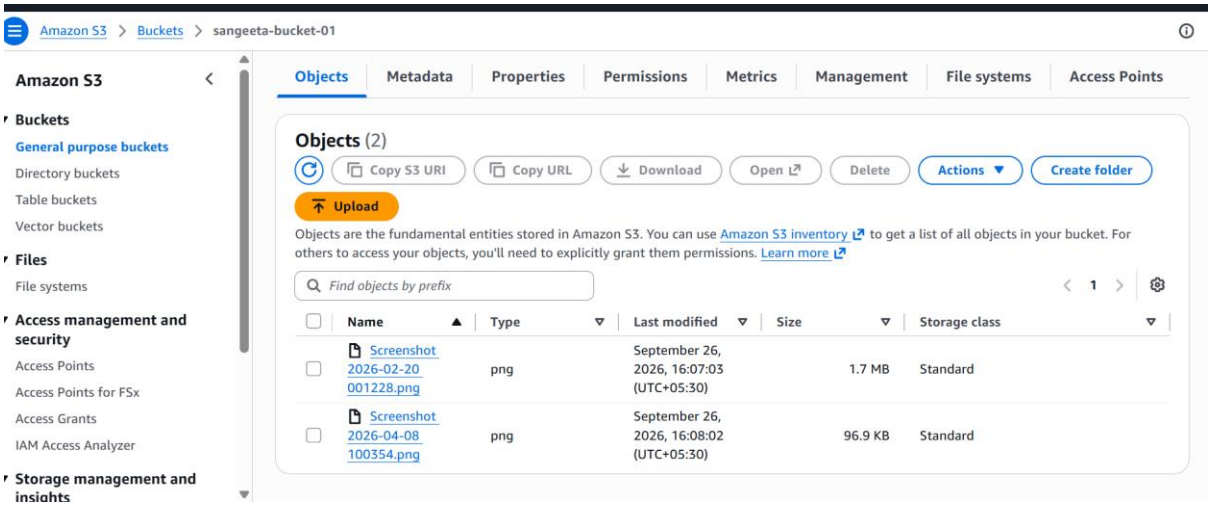
Use your administrator/root account.

Open S3 → Buckets → sangeeta-bucket-01.

Upload two small, non-sensitive files:

approved-file.txt

restricted-file.txt



Part 2: Tag only the approved object

1. Click approved-file.txt.
2. Open the Properties tab.
3. Scroll to Tags.
4. Click Edit or Add tag.

This XML file does not appear to have any style information associated with it. The document tree is shown below.


```
<Error>
  <Code>AccessDenied</Code>
  <Message>User: arn:aws:iam::353362990718:user/Ben is not authorized to perform: s3:GetObject on resource: "arn:aws:s3:::sangeeta-bucket-01/Screenshot 2026-02-20
  001228.png" with an explicit deny in an identity-based policy: arn:aws:iam::353362990718:policy/Portfolio-S3-Explicit-Deny-Policy</Message>
  <RequestId>2V8XMZ3MAAHGFN3T</RequestId>
  <HostId>5wrGoWjFkPXfvnGp0FJlPuz9+q1/wa5/GDl1Ol3/8Rgh3w+YE4n3sbyC615IjKdF4fEt6YIn2wZ7jIz/Nj3hPEn2s6hlnETF</HostId>
</Error>
```